

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0502

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a delivery Route. Create an	BL1 – Remember	5

	Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5
4	Hospital Management System	BL2 – Understand	5

	Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)		
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. □ A department controls a number of PROJECT. Projects have a unique name, number and a single location. □ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. □ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

*I **CHINTHALA SRAVANTHI / 192525463** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1.

Answer:

UPS Product Tracking Information System

The UPS Product Tracking Information System stores information about shipped items, retail centers, and transportation events to track parcel delivery efficiently.

Entities :

- **Shipped_Item:** Item_No (PK), Weight, Dimensions, Insurance_Amount, Destination, Final_Delivery_Date
- **Retail_Center:** Center_ID (PK), Type, Address
- **Transportation_Event:** Schedule_No (PK), Type, Delivery_Route

Relationships

- Retail Center **Receives** Shipped Item (1:M)
- Shipped Item **Transported Through** Transportation Event (M:N)

ER Diagram:

Retail_Center

|

Receives

|

Shipped_Item

|

Transported_Through

|

Transportation_Event

2.

Answer:

Course Management System

A Course Management System helps universities organize courses, sections, classrooms, and examinations. It minimizes scheduling conflicts and efficiently allocates resources.

Entities

- course (C_No, Name, Department)
- Section (S_No, Enrollment) – Weak Entity
- Room (R_No, Capacity, Building)
- Exam (Time)

Relationships:

Relationships

- Course → Section
- Section → Room
- Section → Exam

Benefits

- Easy classroom scheduling, Avoids timetable clashes.
- Better examination management.

ER Diagram:

Course

|

Has

|

Section

|

Scheduled_In

|

Room

|

Exam

3.

Answer:

University Registrar's Office

The Registrar's Office manages academic records of students, instructors, courses, and enrollments.

Entities

- Student
- Course
- Course Offering
- Instructor

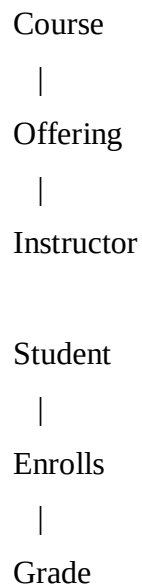
Relationships

- Students enroll in course offerings.
- Instructors teach course offerings.
- Courses have prerequisites.
- Enrollment records grades.

Benefits

- Maintains academic records.
- Generates transcripts.

ER Diagram:



4.

Answer;

Hospital Management System

A Hospital Management System stores information about patients, doctors, tests, treatments, and medical history.

Entities

- Patient (SS#, Name, Insurance)
- Physician (Name, Specialization)
- Test_Log (Test_Name, Date, Time)
- Patient_History

Relationships

- Doctors treat patients.
- Patients undergo tests.
- Test details are stored in the test log.
- Patient history stores previous medical records.

Advantages

- Fast retrieval of patient records.
- Better healthcare management.

ER Diagram:

Physician

|

Treats

|

Patient

|

Has

|

Test_Log

5.

Answer:**Company Project Management System**

A Company Project Management System manages departments, employees, projects, and dependents.

Entities

- Department
- Employee
- Project
- Dependent

Relationships

- Department manages employees.
- Department controls projects.
- Employees work on multiple projects.
- Employees supervise other employees.
- Employees have dependents.

ER Diagram:

Department

|

Controls

|

Project

Employee

|

Works_On

|

Project

Employee

|

Has

|

Depende

6.

Answer:

Schema and Instance

Definition of Schema

A schema is the logical design or blueprint of a database that defines tables, attributes, keys, and relationships.

Definition of Instance

An instance is the actual data stored in the database at a particular moment.

Types of Schema

Logical Schema

- Defines tables and relationships.

Physical Schema

- Describes how data is stored on storage devices.

View Schema

- Shows only the required data to a particular user.

Difference

Schema	Instance
Structure of database	Actual data
Rarely changes	Changes frequently
Created by designer	Updated by users

7.

Answer:

Bank Management System

A Bank Management System stores customer, account, loan, and branch information.

Entities

- Customer
- Account
- Loan
- Branch

Relationships

- Customer owns account.
- Branch maintains account.
- Customer applies for loan.
- Branch approves loan.

Advantages

- Secure customer records.
- Easy transaction processing.
- Better loan management.
- Fast account maintenance.

8.

Answer:

Extended ER (EER) Model

Definition:

The Extended Entity Relationship (EER) Model extends the ER model by supporting inheritance and advanced modeling features.

Concepts

Generalization

- Combines multiple entities into one superclass.

Specialization

- Creates subclasses from a superclass.

Aggregation

- Treats a relationship as an entity for another relationship.

Advantages

- Better representation of complex systems.
- Reduces redundancy.
- Improves database design.
- Supports inheritance.

9.

Answer:

Hospital EER Diagram Superclass

- Person

Subclasses

- Doctor
- Patient

Weak Entity

- Test_Record
- Prescription

Relationships

- Doctor treats patient.
- Patient has test records.
- Doctor prescribes medicines.
- Patient visits doctor.

Advantages

- Better organization.
- Less data duplication.
- Easy maintenance.
- Supports inheritance.

10.

Answer:

Database Administrator (DBA)

Definition:

A **Database Administrator (DBA)** is responsible for installing, maintaining, securing, and managing the database system.

Responsibilities:

1. Install and configure the DBMS.
2. Create and maintain databases.
3. Manage users and permissions.
4. Perform backup and recovery.

5. Ensure security and optimize performance.

Conclusion:

A DBA ensures that the database remains secure, reliable, and available for users.

